

MS-2000 Parts and Materials

Design A, Pan and Tilt Camera Turret. Design locked Sep 16, 2026. Rev 2: USB-C power bank (MOC-004).

Links checked
Sep 15 to 16, 2026
18 line items to buy

\$223.53

Primary order, 4 stores: DFRobot \$152.70 + DigiKey \$37.24 + Adafruit \$9.75 + Walmart \$23.84. Shipping and tax extra.

~\$253.53

All-in with about \$30.00 of AAA batteries, foam board, rod, strap, and craft supplies from local stores.

\$182.85

If every backup were used instead (Adafruit \$94.65, DigiKey \$73.80, DFRobot \$11.90), not counting backups with no listed price (laser, bank, barrel cable).

How to read this

- Every part has a **primary** and a **backup**. Buy the primary. Use the backup only if the primary is out of stock or late.
- Primary uses **four stores**: DFRobot (the micro:bit, camera, board, and servos all come from one ecosystem), DigiKey (the laser and small parts), Adafruit (power cable, switch, and charge port), and Walmart (the power bank, also at Target and Micro Center).
- Backups use Adafruit and DigiKey, plus DFRobot for the backup board and Target, Newark, or Mouser for the power parts.
- Every link was opened on the dates above. Prices are what each site showed.
- Amazon is left out on purpose: its listings change sellers and cannot be checked reliably.

Check before and after ordering

- **DigiKey prices marked *** come from DigiKey's listing on Octopart; confirm in the cart.
- **DigiKey stock is thin** on the 4-wire cable (16). Order it with everything else.
- **DFRobot** ships from its US warehouse by UPS (3 to 7 days to Texas). Items not in the US warehouse ship from China in 10 to 17 business days with duties prepaid.
- **On arrival**: check whether the Xia mi powers the micro:bit, and which 5 V port the servos run best on. With both servos moving and the laser on, the 5 V pins must read at least 4.6 V on the power bank; if not, feed the board's USB 5 V input instead. Measure the bank, the power switch, and the charge port before printing the base.
- **Laser**: read the Class II label on the module before first power-up.

Brains and control

01 micro:bit V2

Qty 2

One runs the turret, one lives in the mosquito wand. They talk by radio, and the wand one records the data.

PRIMARY	DFRobot MBT0039 In stock	\$45.80
	micro:bit V2	2 × \$22.90
	dfrobot.com/product-2125.html	

BACKUP	Adafruit 4781 In stock	\$35.90
	BBC micro:bit v2	2 × \$17.95
	adafruit.com/product/4781	

Board only. Uses micro-USB for code; a cable is almost certainly on hand.

02 Expansion board (turret)

Qty 1

Holds the turret micro:bit. 5 V ports for the servos and speaker, a 5 V HuskyLens port, a relay that switches the laser, and an external power input.

PRIMARY	DFRobot MBT0042 In stock	\$29.90
	Xia mi Multi-functional Expansion Board for micro:bit V2	
	dfrobot.com/product-2371.html	

BACKUP	DFRobot DFR0548 In stock	\$11.90
	micro:Driver expansion board (8 servo, 4 motor)	
	dfrobot.com/product-1738.html	

Xia mi: HuskyLens port (5 V I2C), 5 V GPIO on P0 P1 P2 P8, relay and 4 motor channels, 5 to 12 V external input, MakeCode extension DFRobot/pxt-DFRobot_xia_mi_Board. Backup board: servos and motor channels confirmed; its I2C port voltage for the HuskyLens is not documented.

Camera and aiming

03 HuskyLens AI camera

Qty 1

Learns the fake mosquito and reports where it is on screen.

PRIMARY	DFRobot SEN0305 In stock	Gravity: HuskyLens K210 AI Camera	\$34.90
dfrobot.com/product-1922.html			
BACKUP	DigiKey 1738-SEN0305-ND 68 in stock*	DFRobot SEN0305 (HuskyLens)	\$54.90
digkey.com/en/products/detail/dfrobot/SEN0305/10529331			

Comes with a 4-pin Gravity cable, two brackets, and M3 screws and nuts for its own mounting. MakeCode extension: DFRobot/pxt-DFRobot_HuskyLens.

04 Micro servo

Qty 3

Pan, tilt, and one spare.

PRIMARY	DFRobot SER0049 In stock	9g 180 degree Clutch Servo (damage-proof)	\$14.70
dfrobot.com/product-2120.html			
BACKUP	Adafruit 169 In stock	Micro Servo, TowerPro SG92R	\$17.85
adafruit.com/product/169			

Clutch servo slips instead of stripping gears when a hand grabs the turret. 4.8 to 6 V. Price is the 3+ rate (\$5.00 each for 1 or 2).

Laser and hit sensing

05 Class 2 laser module, under 1 mW

Qty 1

The laser. Maker-labeled Class II.

PRIMARY	DigiKey VLM-650-03-LPT-ND 1,372 in stock*	Quarton VLM-650-03 LPT, 650 nm, <1 mW, 7 mm	\$25.03
digkey.com/en/products/detail/quarton-inc/VLM-650-03-LPT/1010454			
BACKUP	DigiKey VLM-650-22 LPT Ships today	Quarton VLM-650-22 LPT, 650 nm, 0.5 to 0.9 mW, 4 mm	at checkout
digkey.com/en/products/detail/quarton-inc/VLM-650-22-LPT/8636974			

Quarton manuals confirm both LPT versions are Class II, under 1 mW. Not sold by Adafruit, DFRobot, or Mouser. Do not substitute a 5 mW hobby laser (Class IIIa).

06 Laser arm switch

Qty 1

Cuts power to the laser only. Must be on before the laser can fire at all.

PRIMARY	DigiKey Adafruit 805 Price not shown	Breadboard-friendly SPDT slide switch	\$0.95
digkey.com/en/products/detail/adafruit-industries-llc/805/26735538			
BACKUP	Adafruit 805 In stock	Breadboard-friendly SPDT Slide Switch	\$0.95
adafruit.com/product/805			

DigiKey lists this part but did not show a price; \$0.95 is Adafruit's price. Confirm in the cart, or add it to an Adafruit order.

07 Light sensor

Qty 3

One inside each mosquito body. Feels the laser dot.

PRIMARY **DigiKey** 1528-2748-ND 321 in stock*
Adafruit ALS-PT19 analog light sensor breakout
[digikey.com/en/products/detail/adafruit-industries-llc/2748/5775538](https://www.digikey.com/en/products/detail/adafruit-industries-llc/2748/5775538)

\$7.50
3 × \$2.50

BACKUP **Adafruit** 2748 In stock
Adafruit ALS-PT19 Analog Light Sensor Breakout
adafruit.com/product/2748

\$7.50
3 × \$2.50

2.5 to 5.5 V, analog output. Cheaper backup: Adafruit 161 photocell (\$0.95, needs a 10k resistor).

08 Red LED, 5 mm

Qty 10

Mosquito eyes that flash on a hit (2 per mosquito, plus spares).

PRIMARY **DigiKey** 754-1264-ND 145,526 in stock*
Kingbright WP7113ID red diffused 5 mm
[digikey.com/en/products/detail/kingbright/WP7113ID/1747663](https://www.digikey.com/en/products/detail/kingbright/WP7113ID/1747663)

\$1.45
10 × \$0.14

BACKUP **Adafruit** 297 In stock
Super Bright Red 5mm LED (25 pack)
adafruit.com/product/297

\$8.00

Price is the 10-piece rate.

09 220 ohm resistor

Qty 10

One per LED.

PRIMARY **DigiKey** 220QBK-ND 77,901 in stock*
Yageo CFR-25JB-52-220R, 1/4 W
[digikey.com/en/products/detail/yageo/CFR-25JB-52-220R/1295](https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-220R/1295)

\$0.36
10 × \$0.036

BACKUP **Adafruit** 2780 In stock
Through-Hole Resistors, 220 ohm (25 pack)
adafruit.com/product/2780

\$0.75

Price is the 10-piece rate.

10 Thin 4-wire cable, 1 m

Qty 1

Runs down the fishing line to the sensor and eyes: power, ground, sensor, LEDs.

PRIMARY **DigiKey** 1528-2694-ND 16 in stock*
Adafruit 3891 silicone ribbon cable, 4 wire, 28 AWG, 1 m
[digikey.com/en/products/detail/adafruit-industries-llc/3891/9603617](https://www.digikey.com/en/products/detail/adafruit-industries-llc/3891/9603617)

\$1.95

BACKUP **Adafruit** 3890 In stock
Silicone ribbon cable, 10 wire, 28 AWG, 1 m (peel off 4)
adafruit.com/product/3890

\$3.95

Low stock at DigiKey: order early.

Sound

11 Speaker module

Qty 1

Laser pew, cannon boom, lock-on beep, victory sound. Volume knob.

PRIMARY	DFRobot FIT0449 In stock Gravity: Digital Speaker Module dfrobot.com/product-1401.html	\$6.00
BACKUP	Adafruit 2130 + 3923 In stock PAM8302 2.5 W mono amp (\$3.95) + mini oval 8 ohm speaker (\$1.95) adafruit.com/product/2130	\$5.90

Plugs into the Xia mi P0 port (5 V); MakeCode plays sound on P0. Backup speaker page: adafruit.com/product/3923 (solder the wires to the amp).

Power

12 Turret power bank, USB-C

Qty 1

Powers the Xia mi board: servos, camera, speaker, laser. Steady 5 V, stays inside the base and charges through the side port (change MOC-004, Sep 16).

PRIMARY	Walmart A1112 In stock Anker 321 Power Bank (PowerCore 5K), 5,200 mAh, USB-A + USB-C walmart.com/ip/Anker-5K-Power-Bank-12W-USB-C-USB-A-Port/2335161189	\$23.84
BACKUP	Target A-86373147 Check store Same bank (Anker 5000mAh PowerCore 5K 12W 1A1C) target.com/p/anker-5000mah-power-bank-powercore-5k-12w-1a1c-black/-/A-86373147	at checkout

97 x 45.8 x 22 mm, 118 g, 5 V up to 2.4 A. Same bank as Nolan's Dusty. The turret draws enough that the bank never shuts itself off; do not use its trickle mode (double press). Also sold at Micro Center Houston.

13 USB to barrel jack cable

Qty 1

Bank USB-A port to the power switch, then the Xia mi DC jack.

PRIMARY	Adafruit 2697 In stock USB to 2.1mm Male Barrel Jack Cable, 22AWG, 1 m adafruit.com/product/2697	\$2.75
BACKUP	Newark 2697 Check stock Adafruit 2697 (same cable) newark.com/adafruit/2697/usb-to-2-1mm-male-barrel-jack/dp/77Y5319	at checkout

Center positive, 5.5/2.1 mm. Coil the extra length inside the base. A 90 degree USB-A adapter (spare from Dusty's pair) turns the plug so it fits in front of the bank.

14 Power switch

Qty 1

Main on/off. Plugs between the barrel cable and the Xia mi jack; the rocker pokes through the right wall.

PRIMARY	Adafruit 1125 In stock In-line power switch for 2.1mm barrel jack, 2 A adafruit.com/product/1125	\$2.50
BACKUP	Mouser 1125 Check stock Adafruit 1125 (same switch) mouser.com/c/?q=adafruit%201125	\$2.50

Body size is not published: measure it before printing the base.

15 USB-C charge port

Qty 1

Panel USB-C socket in the right wall. Its cable plugs into the bank, so a phone charger charges the turret from outside.

PRIMARY	Adafruit 6069 In stock USB C Small Round Panel Mount Extension Cable adafruit.com/product/6069	\$4.50
BACKUP	Adafruit 4218 In stock USB C Round Panel Mount Extension Cable, 30 cm (needs a 21.5 mm hole) adafruit.com/product/4218	\$9.95

Fits a 12 to 18 mm hole; the base has 14 mm. Also listed at DigiKey (6069). Switch off before charging; plugging in can blink the power and reset the micro:bit.

16 micro:bit battery pack

Qty 2

Powers the wand micro:bit, and the turret micro:bit if the board does not.

PRIMARY	DFRobot FIT0625 In stock 2xAAA battery holder with cover and switch, PH2.0 dfrobot.com/product-1855.html	\$2.50 2 × \$1.25
BACKUP	Adafruit 4193 In stock 2xAA battery holder with on/off switch and JST PH adafruit.com/product/4193	\$3.90 2 × \$1.95

The micro:bit battery socket is JST PH 2.0 mm. Adafruit's 2xAAA version (4191) is out of stock, so the backup is 2xAA.

Wiring and hardware

17 M2 screw, nut, and standoff kit

Qty 1

Every printed part: base, turret head, wand handle, sensor pod, and backdrop parts.

PRIMARY	DFRobot FIT0665 In stock Metal screw, nut and mounting kit, M2, 320 pcs dfrobot.com/product-2023.html	\$12.90
BACKUP	DigiKey 1738-FIT0665-ND 321 in stock* DFRobot FIT0665 (same kit) digikey.com/en/products/detail/dfrobot/FIT0665/12324933	\$12.90

Prints use M2 clearance holes (2.2 mm) and M2 nut traps. Exceptions: HuskyLens uses its own M3 kit; servos use their own horn screws.

18 Gravity 3-pin cables (10 pack)

Qty 1

Servo, speaker, and laser leads to the board, plus spares.

PRIMARY	DFRobot FIT0031 In stock Gravity: Analog Sensor Cable, 30 cm (10 pack) dfrobot.com/product-128.html	\$6.00
BACKUP	DigiKey 1738-1031-ND 111 in stock* DFRobot FIT0031 (same cables) digikey.com/en/products/detail/dfrobot/FIT0031/6588453	\$6.00

Order sheets

DFRobot cart (primary)

#	SKU	PART	QTY	PRICE
01	MBT0039	micro:bit V2	2	\$45 . 80
02	MBT0042	Expansion board (turret)	1	\$29 . 90
03	SEN0305	HuskyLens AI camera	1	\$34 . 90
04	SER0049	Micro servo	3	\$14 . 70
11	FIT0449	Speaker module	1	\$6 . 00
16	FIT0625	micro:bit battery pack	2	\$2 . 50
17	FIT0665	M2 screw, nut, and standoff kit	1	\$12 . 90
18	FIT0031	Gravity 3-pin cables (10 pack)	1	\$6 . 00
Subtotal			\$152 . 70	

DFRobot has offered free US shipping over \$50 as a limited-time deal; check at checkout.

DigiKey cart (primary)

#	DIGIKEY / PART	PART	QTY	PRICE
05	VLM-650-03-LPT-ND	Class 2 laser module, under 1 mW	1	\$25 . 03
06	Adafruit 805	Laser arm switch	1	\$0 . 95
07	1528-2748-ND	Light sensor	3	\$7 . 50
08	754-1264-ND	Red LED, 5 mm	10	\$1 . 45
09	220QBK-ND	220 ohm resistor	10	\$0 . 36
10	1528-2694-ND	Thin 4-wire cable, 1 m	1	\$1 . 95
Subtotal			\$37 . 24	

Resistors and LEDs are sold loose; type the quantity in the cart.

Adafruit cart (primary)

#	PRODUCT ID	PART	QTY	PRICE
13	2697	USB to barrel jack cable	1	\$2 . 75
14	1125	Power switch	1	\$2 . 50
15	6069	USB-C charge port	1	\$4 . 50
Subtotal			\$9 . 75	

Add the arm switch (805) here too if DigiKey will not price it.

Walmart (primary)

#	MODEL	PART	QTY	PRICE
12	A1112	Turret power bank, USB-C	1	\$23 . 84
Subtotal			\$23 . 84	

Store pickup works. Charge it fully when it arrives.

From local stores or already on hand

ITEM	AMOUNT	WHERE
AAA batteries	4 plus 4 spares	Grocery store
Stuffed mosquito toy (the target)	1	Toy store or online
Short fishing rod or 3/8" dowel, clear 10 lb line	1	Sporting goods or hardware store
Black foam board, 20x30 in, 3/16" thick	1	Craft store
1/4" wood dowel	1	Craft or hardware store
Small zip ties and hook-and-loop tape	1 pack each	Hardware store
PLA filament, any color plus a little white	About 310 g + 11 g white	On hand (check for white)
Solder, heat-shrink tubing, hot glue	Small amounts	On hand
Micro-USB data cables	2	Probably on hand
90 degree USB-A adapter	1	Spare from Dusty's pair (backup: Walmart 709931832)
USB-C phone charger and cable	1	On hand
Hook-and-loop strap, 20 mm	1	Hardware store (holds the bank)
Paper data sheets, printed	15 per test	Print at home

What the design locks in

- **Design A:** pan and tilt camera turret, HuskyLens on top, laser below it, boresighted.
- **Printed tilt stop** plus a code limit: cannot aim above the backdrop.
- **Laser path:** 5 V port, through the arm switch, through the board relay, to the laser.
- **Ports:** HuskyLens on the 5 V HuskyLens port; pan servo P1, tilt servo P2, speaker P0 (all 5 V); P8 spare.
- **Turret power:** USB-C power bank on the base floor, bank to power switch to Xia mi DC jack; USB-C charge port in the right wall.
- **Wand:** second micro:bit in a printed handle, 2xAAA pack, 4-wire cable down the line to the mosquito (sensor + LED eyes).
- **Fasteners:** M2 everywhere in printed parts; M3 only for the HuskyLens.

If the primary board gives trouble

- Switch to the **micro:Driver** (backup #02): servos on its servo headers, laser on motor channel M1, power from the same power bank (5 V is inside its 3.5 to 5.5 V range).
- Its I2C port voltage is not documented. Test the HuskyLens on it before relying on it; if it browns out, power the HuskyLens from the bank's second (USB-C) port.
- The printed base has room for either board (Xia mi 57 × 87 mm is the larger).